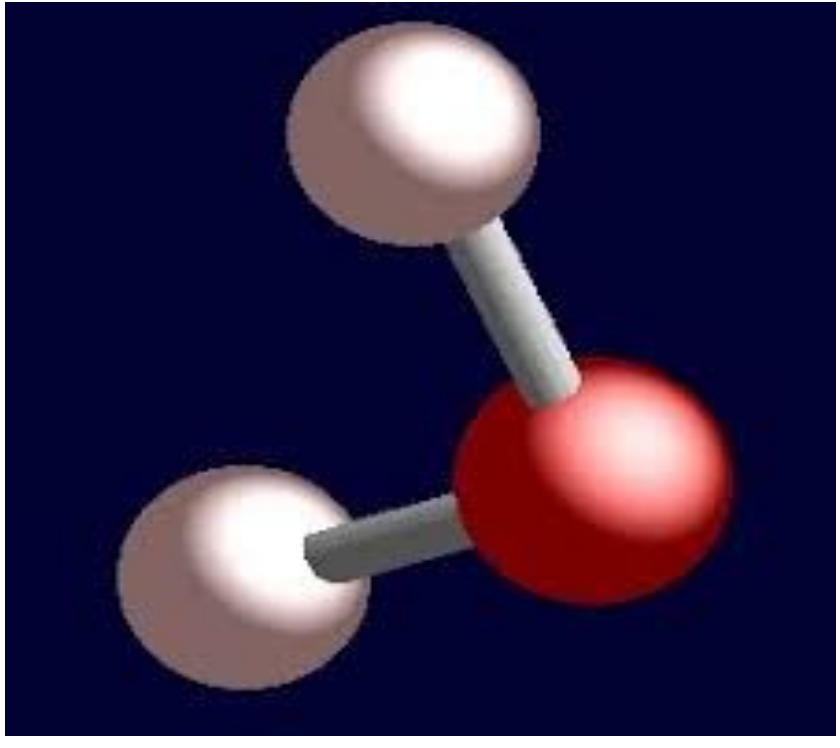
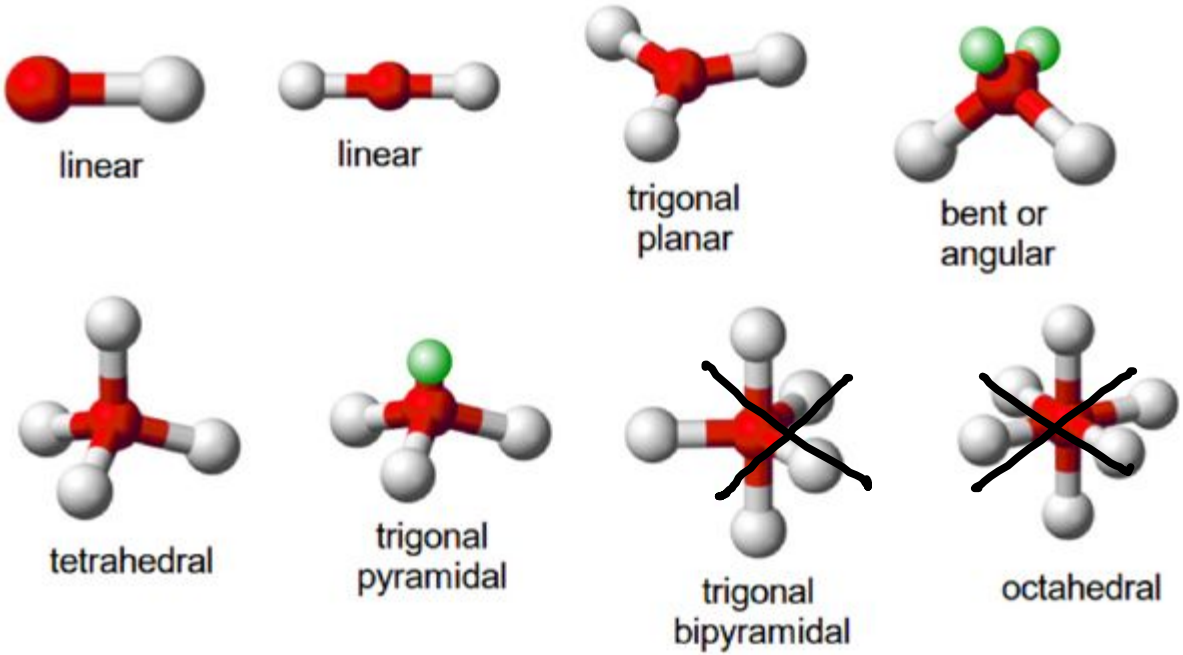
3D Structure & Polarity



VSEPR Theory

- VSEPR - Valence Shell Electron Pair Repulsion
- Electrons orient themselves in order to minimize the repulsive forces.

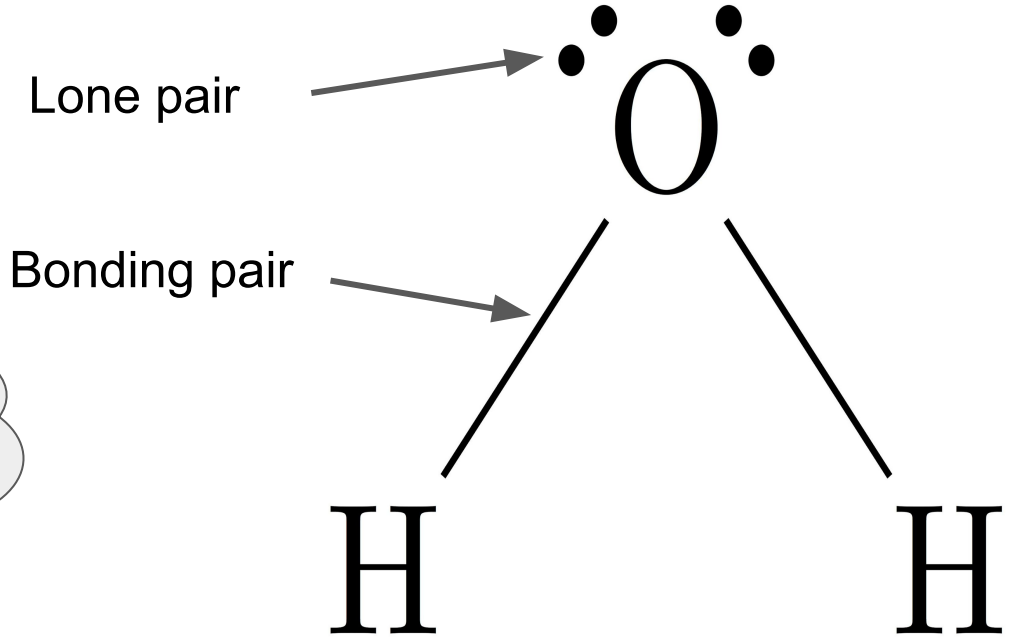


VSEPR Theory

Types of Electron Pairs

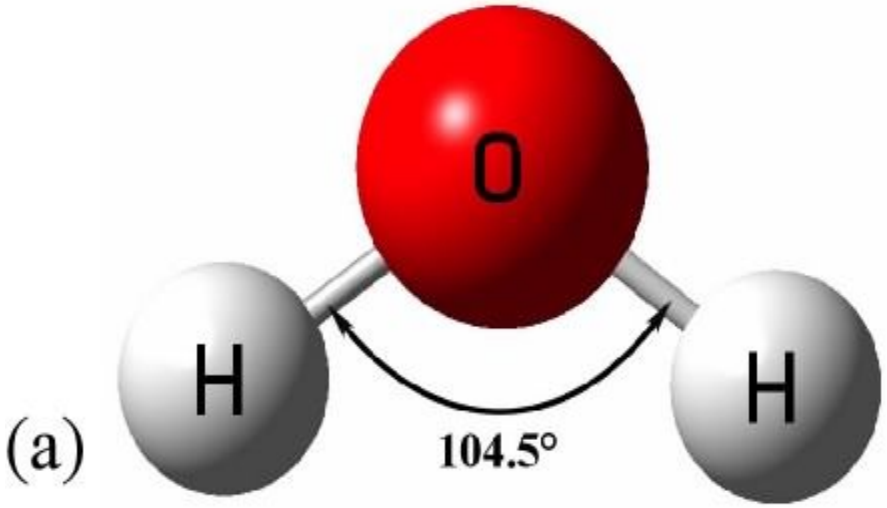
- Bonding pairs - electrons that form bonds
- Lone pairs - non-bonding electrons

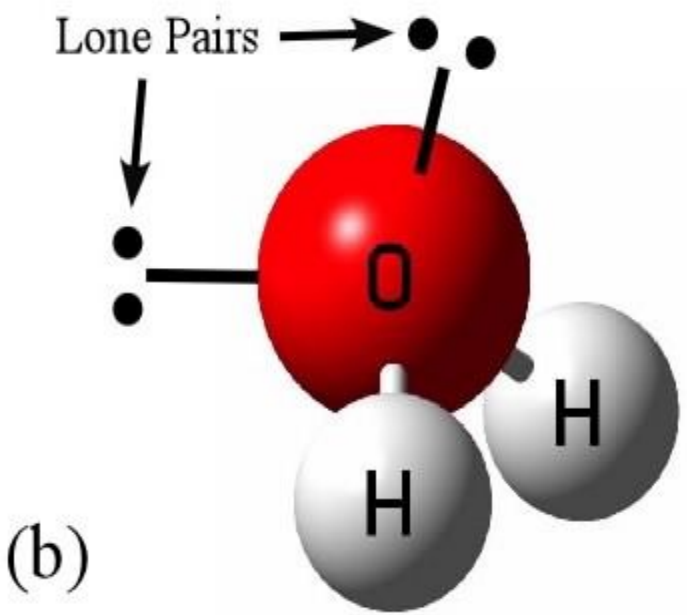
Lone pairs repel more strongly than bonding pairs!



VSEPR Theory

- Lone pairs reduce the bond angles between atoms.





Determining Molecular Shape

- Draw the Lewis Structure
- Tally up electron pair around the central atom
  - Find bonding pairs and lone pairs
  -  double/triple count as one pair
- Shape is determined by the number of bonding and lone pairs of electrons.

You will need to know
the 7 common shapes
and their bond angles

VSEPR Chart

Bond Pair (attached to the central atom)	Lone Pair (attached to the central atom)	Shape	Angle
1	0	linear	180°
2	0	linear	180°
3	0	Trigonal planar	120°
2	1	Bent	<120°
4	0	Tetrahedral	109.5°
3	1	Trigonal Pyramidal	107°
2	2	Bent	104.5°

N_2 $2(5) = 10 - 2 = 8 - 8 = \emptyset$

Valence Electrons 10 Ve⁻

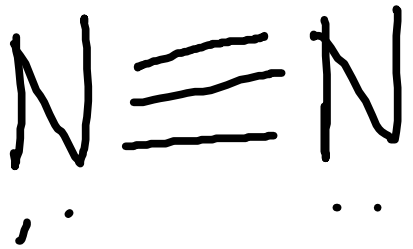
Bond pair 1

Lone pair 0

Shape linear

Angle 180°





HCN $1+4+5=10-4=6-6=\emptyset$

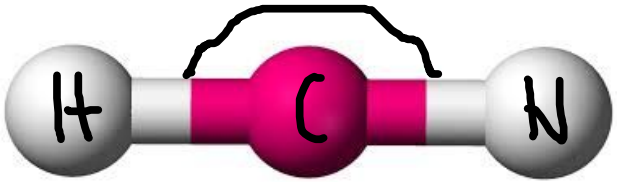
Valence Electrons 10 ve^-

Bond pair 2

Lone pair 0

Shape linear

Angle 180°



$H-C\equiv N:$

$[CO_3]^{-2}$ $4+3(6)=22+2=24$

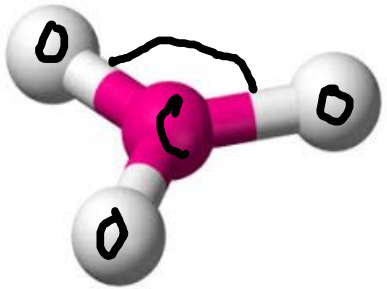
Valence Electrons 24 ve⁻

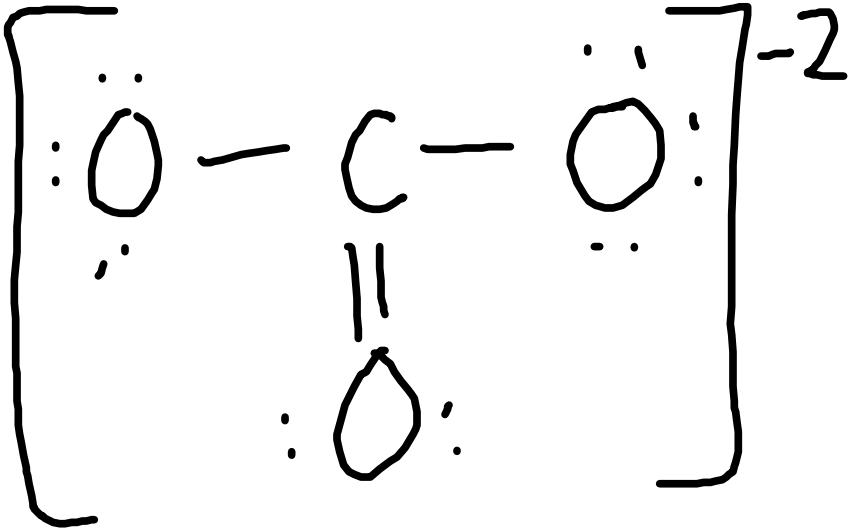
Bond pair 3

Lone pair 0

Shape trigonal planar

Angle 120°





SO_2 $6 + 2(6) = 18 - 4 = 14 - 12 = 2 - 2 = \emptyset$

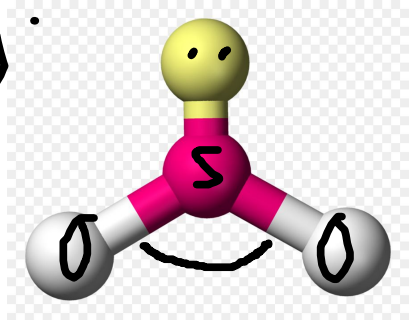
Valence Electrons 18ve⁻

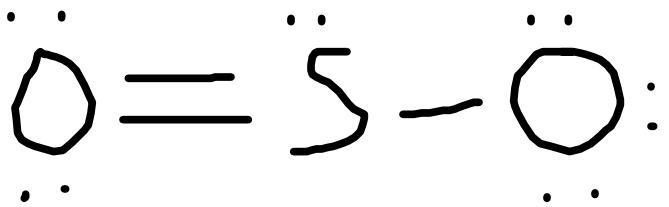
Bond pair 2

Lone pair 1

Shape bent

Angle $< 120^\circ$





CH_4 $4+4(1)=8-8=\emptyset$

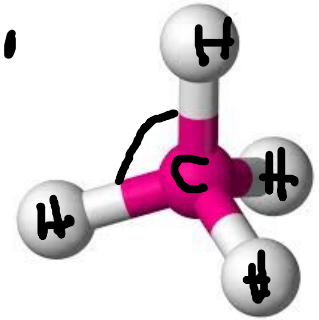
Valence Electrons 8 ve-

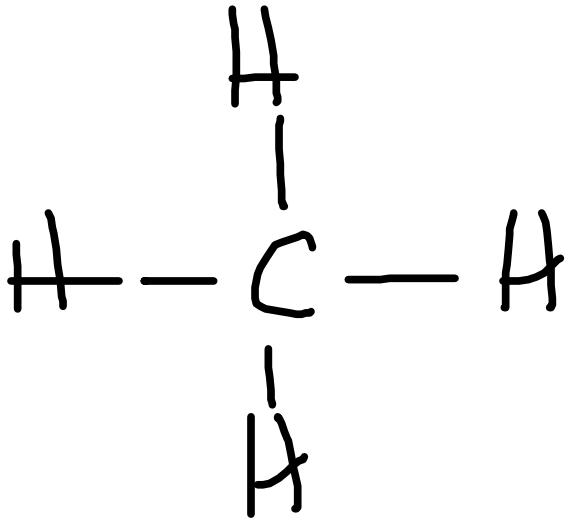
Bond pair 4

Lone pair 0

Shape tetrahedral

Angle 109.5°





NH_3 $5+3(1)=8-6=2$

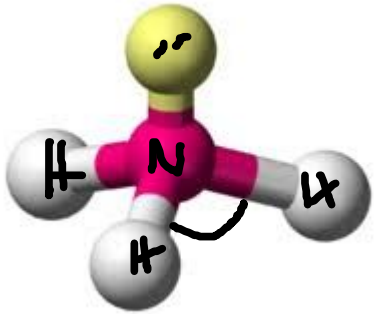
Valence Electrons 8 ve⁻

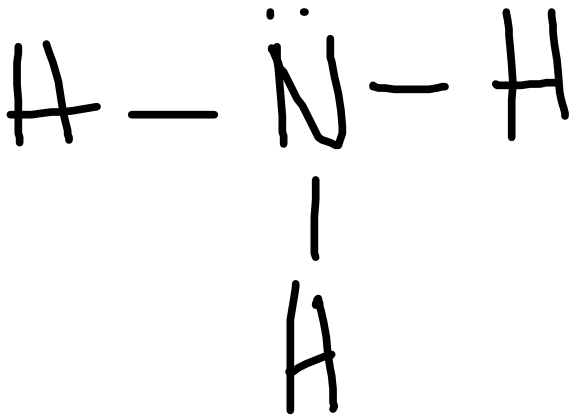
Bond pair 3

Lone pair 1

Shape trigonal pyramidal

Angle 107°





H_2O $2(1)+6=8-4=4-4=0$

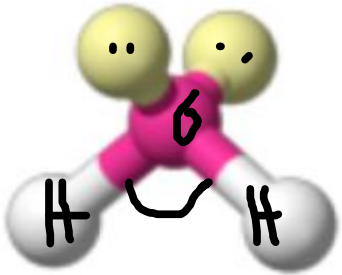
Valence Electrons 8 ve^-

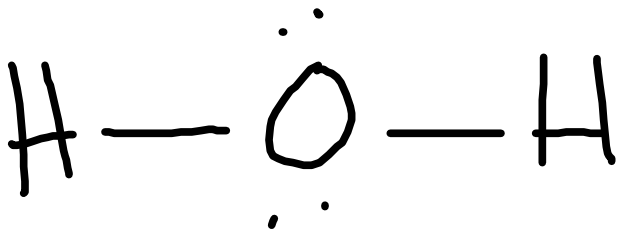
Bond pair 2

Lone pair 2

Shape bent

Angle 104.5°





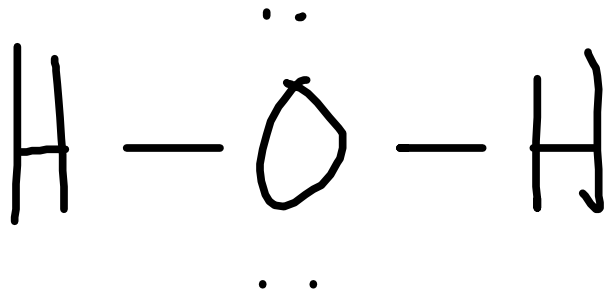
Molecular Polarity

- How to determine if a molecule is polar or non-polar
  - The molecule is polar IF…
    - There are different outside atoms ***OR***
    - There are lone pair(s) of electrons around the central atom
  - Otherwise the molecule is non-polar

Molecular Polarity

H_2O

Polar or Non-polar?



Molecular Polarity

CBr_4

Polar or Non-polar?

